Exact Owner Subtraction and Direct Analytic Closure

Analytic ledger replacement, Packet F

Abstract

The executable exact ledger checks a 9×7 truth table when the accepted lower and staircase regions are subtracted from the Round–19 residual. This note replaces all sixty-three state checks and the five additional interface checks by one set-theoretic calculation. The surviving set is then placed directly inside the closed domain of the all-frequency retained-remainder theorem. Every open and closed face is retained explicitly, and no artificial frequency interface is introduced.

1 Sets and inherited owners

Let

$$0 < \rho_* < \rho_0 < \rho_c < \rho_{\mathrm{opt}}, \qquad \rho_{\mathrm{opt}} := \frac{39}{50},$$

and let $U(\rho)$, $k_6(\rho)$, and $k_{11}(\rho)$ be the upper owner and the two staircase walls from the main proof. Put

$$\begin{aligned} D_{\mathrm{low}} :=& \left\{\rho_* < \rho \le \rho_0,\ (2\rho)^{-1} < K < U(\rho)\right\} \\ & \cup \{\rho_0 < \rho < \rho_c,\ d < K < U(\rho)\}\,, \end{aligned} \tag{1}$$

$$D_{\mathrm{high}} := \{\rho_c \le \rho < \rho_{\mathrm{opt}},\ k_6(\rho) < K < U(\rho)\}\,, \tag{2}$$

where $d = (2\rho_0)^{-1}$, and define

$$D_{19} := D_{\mathrm{low}} \,\dot{\cup}\, D_{\mathrm{high}}. \tag{3}$$

The dot is justified by the disjoint ratio ranges.

The already-proved inclusive owners used in the subtraction are:

$$L_{\mathrm{own}} := D_{\mathrm{low}}, \tag{4}$$

$$S_{\mathrm{own}} := \{\rho_c \le \rho < \rho_{\mathrm{opt}},\ k_6(\rho) < K \le k_{11}(\rho)\}\,, \tag{5}$$

The lower owner includes the complete $\rho = \rho_0$ fibre. The staircase includes its upper wall $K = k_{11}(\rho)$. The optical owner from the main proof has already restricted the high component of D_{19} to $\rho < \rho_{\mathrm{opt}}$. The residual also excludes $K = U(\rho)$.

2 The subtraction

Proposition 1 (Exact residual identity)**.** *One has*

$$\boxed{D_{19} \setminus (L_{\mathrm{own}} \cup S_{\mathrm{own}}) = D_{20},} \tag{6}$$

where

$$D_{20} := \{\rho_c \le \rho < \rho_{\mathrm{opt}},\ k_{11}(\rho) < K < U(\rho)\}\,. \tag{7}$$

Proof. Since $L_{\text{own}} = D_{\text{low}}$ and the union in (3) is disjoint,

$$D_{19} \setminus L_{\text{own}} = D_{\text{high}}. \tag{8}$$

Inside D_{high}, the lower frequency inequality is strict: $k_6(\rho) < K$. Removing the inclusive staircase (5) therefore gives exactly

$$D_{\text{high}} \setminus S_{\text{own}} = \{\rho_c \le \rho < \rho_{\text{opt}},\ k_{11}(\rho) < K < U(\rho)\}. \tag{9}$$

There is no missing equality case: $K = k_{11}(\rho)$ belongs to S_{own}, while $K = U(\rho)$ never belonged to D_{19}. The face $\rho = \rho_{\text{opt}}$ was already removed when the optical theorem was applied in forming D_{19}, whereas $\rho = \rho_c$ remains in D_{high}. Thus (9) is precisely (7). □

3 The five delicate interfaces

For clarity, the five extra assertions formerly checked separately by the executable ledger are consequences of the displayed sets:

1. At $\rho = \rho_0$, the identity $(2\rho_0)^{-1} = d$ makes the prospective second lower fibre empty; the first set in (1) owns the full equality fibre, and L_{own} removes it exactly once.
2. The face $\rho = \rho_c$ belongs to D_{high}. It is not an optical face, so points with $k_{11}(\rho_c) < K < U(\rho_c)$ remain in D_{20}.
3. The face $\rho = \rho_{\text{opt}} = 39/50$ belongs to the optical theorem and is already absent from D_{19}, hence from D_{20}.
4. The frequency face $K = k_{11}(\rho)$ belongs to S_{own}, whereas $K = U(\rho)$ is excluded already by the strict inequality in D_{19}.
5. If $\rho_c < \rho < \rho_{\text{opt}}$ and $k_{11}(\rho) < K < U(\rho)$, the point is in D_{high} but in neither L_{own} nor S_{own}; hence it survives, and every surviving point has exactly these properties.

4 Direct retained-remainder ownership

Let

$$\mathcal{R}_{\text{rr}} := \{(\rho, K) : \rho_c \le \rho \le \rho_{\text{opt}}, \quad K \ge k_{11}(\rho)\}. \tag{10}$$

This is the closed domain of the all-frequency retained-remainder theorem proved by the structural, angular-block, and scalar-positivity modules. In the notation of the main proof, that theorem states

$$(\rho, K) \in \mathcal{R}_{\text{rr}} \quad \Longrightarrow \quad N_D(A_{\rho,1}, K^2) < W(\rho, K). \tag{11}$$

Proposition 2 (Direct ownership of the exact residual)**.** *One has*

$$\boxed{D_{20} \subset \mathcal{R}_{\text{rr}}}, \qquad D_{20} \setminus \mathcal{R}_{\text{rr}} = \varnothing. \tag{12}$$

Consequently the strict Pólya inequality holds at every point of D_{20}.

Proof. If $(\rho, K) \in D_{20}$, then

$$\rho_c \le \rho < \rho_{\mathrm{opt}} \quad \text{and} \quad k_{11}(\rho) < K < U(\rho).$$

The first pair of inequalities implies $\rho_c \le \rho \le \rho_{\mathrm{opt}}$, and the second implies $K \ge k_{11}(\rho)$. Hence $(\rho, K) \in \mathcal{R}_{\mathrm{rr}}$, which proves (12); the final assertion follows from (11).

The inclusion does not change any inherited owner. The face $\rho = \rho_{\mathrm{opt}}$ and the frequency face $K = k_{11}(\rho)$ are outside D_{20}, although the stronger theorem also proves them. The face $K = U(\rho)$ remains excluded from D_{20}, while $\rho = \rho_c$ remains included subject to the two strict frequency inequalities. There is no additional frequency face. $\square$

Ledger rows replaced. Proposition 1 replaces all $9 \cdot 7 = 63$ state evaluations in the executable family `verify_exact_subtraction`. The five items above replace its five separately asserted interface predicates. Proposition 2 supplies the final residual closure by one set inclusion, without a frequency classifier or executable truth table.